Final Project Databases

The topic I wanted to research for my final in this database is Academic Success Patterns. With questions underlined, I have the queries below them, and at the end, the answer to the question.

- <u>Do students who study more than 10 hours per week earn higher final grades in their courses?</u>

To answer this question, I wrote 2 queries to show the trend of grades when working more than 10 total study hours and less than 10 total study hours.

```sql
SELECT
AVG(
    CASE
        WHEN cg.final_grade IN ('A', 'A-', 'A+') THEN 4
        WHEN cg.final_grade IN ('B', 'B-', 'B+') THEN 3
        WHEN cg.final_grade IN ('C', 'C-', 'C+') THEN 2
        WHEN cg.final_grade IN ('D', 'D-', 'D+') THEN 1
        WHEN cg.final_grade = 'F' THEN 0
    END
) AS avg_grade_under_10h
FROM course_grades AS cg
INNER JOIN study_habits AS sh
USING (student_id)
WHERE sh.hours_solo + sh.hours_tutoring + sh.hours_group < 10
```

	avg_grade_under_10h
1	2.58130841121495

For this query, it is showing the average grade point average for students who studied less than 10 hours, which is around 2.5 GPA. This does not take into account the grade's value, or whether it is a plus or minus grade; it is simply the letter grade that the students receive.

```sql
SELECT
AVG(
    CASE
        WHEN cg.final_grade IN ('A', 'A-', 'A+') THEN 4
        WHEN cg.final_grade IN ('B', 'B-', 'B+') THEN 3
        WHEN cg.final_grade IN ('C', 'C-', 'C+') THEN 2
        WHEN cg.final_grade IN ('D', 'D-', 'D+') THEN 1
        WHEN cg.final_grade = 'F' THEN 0
    END
) AS avg_grade_over_10h
FROM course_grades AS cg
INNER JOIN study_habits AS sh
USING (student_id)
WHERE sh.hours_solo + sh.hours_tutoring +sh.hours_group >= 10
```

	avg_grade_over_10h
1	3.15267175572519

For this query, it is just the exact opposite of the last query. This shows the grade point average of students who study 10 hours or more in total, which is around a 3.15 grade point average. This also only takes into account the letter grade.

With these 2 queries, we have learned that students who study 10 or more hours do, in fact, earn, on average, a higher grade than if they studied less than 10 hours.

- <u>Do group study hours correlate with higher performance in specific majors?</u>

To answer this question, I will use one query that pretty shows the correlation of performance in each major.

GPA under 10hr | GPA over 10hr | GPA group study | GPA tutoring | SQL 9*

```sql
WITH numeric_grades AS (
    SELECT
        student_id,
      CASE
        WHEN final_grade IN ('A', 'A-', 'A+') THEN 4
        WHEN final_grade IN ('B', 'B-', 'B+') THEN 3
        WHEN final_grade IN ('C', 'C-', 'C+') THEN 2
        WHEN final_grade IN ('D', 'D-', 'D+') THEN 1
        WHEN final_grade = 'F' THEN 0
        END AS y
    FROM course_grades
),
joined AS (
    SELECT
        s.major,
        sh.hours_group AS x,
        ng.y
    FROM students s
    JOIN study_habits sh ON s.student_id = sh.student_id
    JOIN numeric_grades ng ON s.student_id = ng.student_id
),
stats AS (
    SELECT
        major,
        COUNT(*) AS n,
        SUM(x) AS sum_x,
        SUM(y) AS sum_y,
        SUM(x*y) AS sum_xy,
        SUM(x*x) AS sum_x2,
        SUM(y*y) AS sum_y2
    FROM joined
    GROUP BY major
)
SELECT
    major,
    (n * sum_xy - sum_x * sum_y) /
    (SQRT((n * sum_x2 - sum_x * sum_x) * (n * sum_y2 - sum_y * sum_y)))
        AS correlation_group_hours_vs_performance
FROM stats
ORDER BY correlation_group_hours_vs_performance DESC;
```

	major	correlation_group_hours_vs_performance
1	Biology	0.145290212617115
2	Computer Science	0.135198596949013
3	Sociology	0.118790544524736
4	Business	0.0091669149022145
5	Psychology	-0.133516320853408
6	Art	-0.136134166745176

With this query, the final result in the “correlation_group_hours_vs_performance” is given in a decimal number. This number represents positive and negative correlation. So if the number is close to 0, it doesn’t have much of a correlation with the major grade, if the number is close to +1.0, it has a positive correlation to the major's grade, and finally, if it's close to -1.0, it has a negative correlation to the major's grade. We can see that, in Psychology and Art majors, group study hours had a very slight negative effect on their grades. We can also see that Business major grades were not affected at all, and Sociology, computer science, and Biology majors' grades were positively affected by the group study hours

- <u>Are students who rely primarily on solo study performing better in upper-level courses?</u>

```sql
SELECT
    s.first_name || ' ' || s.last_name AS student_name,
    cg.final_grade,
    C.course_code
FROM students s
INNER JOIN course_grades AS cg
    ON s.student_id = cg.student_id
INNER JOIN courses AS C
    ON cg.course_id = C.course_id
INNER JOIN study_habits AS sh
    ON s.student_id = sh.student_id
WHERE sh.hours_solo < sh.hours_tutoring + sh.hours_group
  AND (
        C.course_code LIKE '%102'
     OR C.course_code LIKE '%103'
     OR C.course_code LIKE '%104'
  );
```

This query shows the students' grades in the course they took and the final grade they got in that class. These are the students who don't study solo. We consider upper-level classes as anything that is not a 101 class, so we excluded those classes from the results. Looking at the table of the results, the column had 13 F grades with 92 students. That means 14% of the students who don't primarily use solo studying got an F.

```sql
SELECT
    s.first_name || ' ' || s.last_name AS student_name,
    cg.final_grade,
    C.course_code
FROM students s
INNER JOIN course_grades AS cg
    ON s.student_id = cg.student_id
INNER JOIN courses AS C
    ON cg.course_id = C.course_id
INNER JOIN study_habits AS sh
    ON s.student_id = sh.student_id
WHERE sh.hours_solo > sh.hours_tutoring + sh.hours_group
  AND (
        C.course_code LIKE '%102'
     OR C.course_code LIKE '%103'
     OR C.course_code LIKE '%104'
  );
```

This query shows the students' grades in the course they took and the final grade they got in that class. These are the students who primarily study solo. We consider upper-level classes as anything that is not a 101 class, so we excluded those classes from the results. Looking at the table of the results, the column had 29 F grades with 519 students. That means only around 5.5% of the students who primarily use solo studying got an F.

So, looking at the results, we can see that solo students do perform better in high level courses

RESULTS

https://docs.google.com/spreadsheets/d/1cq96MId7iF41aCnSmY0eDi9EEk6VurXjnhUJJwG3XKw/edit?usp=sharing

- How do GPA averages compare for students who use group study vs. tutoring?

To answer this, I used a similar query to the first question.

GPA under 10hr | GPA over 10hr | GPA group study* | GPA tutoring* | SQL 9*

```sql
---Do students who study more than 10 hours per week earn higher final grades in their courses?
SELECT
AVG(
    CASE
        WHEN cg.final_grade IN ('A', 'A-', 'A+') THEN 4
        WHEN cg.final_grade IN ('B', 'B-', 'B+') THEN 3
        WHEN cg.final_grade IN ('C', 'C-', 'C+') THEN 2
        WHEN cg.final_grade IN ('D', 'D-', 'D+') THEN 1
        WHEN cg.final_grade = 'F' THEN 0
    END
) AS avg_grade
FROM course_grades AS cg
INNER JOIN study_habits AS sh
USING (student_id)
WHERE sh.hours_group < sh.hours_tutoring;
```

	avg_grade
1	2.76569037656904

This query is showing the GPA of the student who likes to use group study over tutoring. We do this by just asking it to find the student who has more group study hours than tutoring study hours.

GPA under 10hr | GPA over 10hr | GPA group study* | GPA tutoring* | SQL 9*

```sql
--Do students who study more than 10 hours per week earn higher final grades in their courses?
SELECT
AVG(
    CASE
        WHEN cg.final_grade IN ('A', 'A-', 'A+') THEN 4
        WHEN cg.final_grade IN ('B', 'B-', 'B+') THEN 3
        WHEN cg.final_grade IN ('C', 'C-', 'C+') THEN 2
        WHEN cg.final_grade IN ('D', 'D-', 'D+') THEN 1
        WHEN cg.final_grade = 'F' THEN 0
    END
) AS avg_grade
FROM course_grades AS cg
INNER JOIN study_habits AS sh
USING (student_id)
WHERE sh.hours_group > sh.hours_tutoring ;
```

	avg_grade
1	2.75735294117647

This query is showing the GPA of the student who likes to use tutoring over group study hours. We do this by just asking it to find the student who has more tutoring hours than group study hours.

To answer this question, we can see that the GPAs are very similar, with not much difference being shown.

- <u>Are frequent studiers also students with the highest final grades?</u>

I wrote a query giving a list of the students' total study hours and their GPA.

```sql
--Are frequent studiers also students with the highest final grades?
SELECT
    s.student_id,
    s.first_name || ' ' || s.last_name AS student_name,
    (sh.hours_solo + sh.hours_tutoring + sh.hours_group) AS total_study_hours,
    AVG(
   CASE
        WHEN cg.final_grade IN ('A', 'A-', 'A+') THEN 4
        WHEN cg.final_grade IN ('B', 'B-', 'B+') THEN 3
        WHEN cg.final_grade IN ('C', 'C-', 'C+') THEN 2
        WHEN cg.final_grade IN ('D', 'D-', 'D+') THEN 1
        WHEN cg.final_grade = 'F' THEN 0
    END
) AS avg_grade
FROM students s
JOIN study_habits sh ON s.student_id = sh.student_id
JOIN course_grades cg ON s.student_id = cg.student_id
GROUP BY s.student_id
ORDER BY total_study_hours DESC;
```

RESULT

https://docs.google.com/spreadsheets/d/1NxH006rER9BGfOPwYkLHnUOc3ImuClwm8bPNUVhM0dk/edit?usp=sharing

If we look at the results of this query, we can see that for the higher total study hours, there are more 4.0 and 3.5 GPAs, but if we look at the lower total study hours, then we can see the GPAs steadily lower as the study hours lower. So to answer the question, yes, the more frequently students get higher final grades.

- <u>Which majors or years of study show the widest variation in study strategies?</u>

GPA over 10hr | GPA group study* | GPA tutoring* | SQL 9* | SQL 7* | SQL 9

```sql
-- Which majors or years of study show the widest variation in study strategies?
WITH grade_calc AS (
    SELECT
        s.student_id,
        s.first_name || ' ' || s.last_name AS student_name,
        AVG(
            CASE
                WHEN cg.final_grade IN ('A', 'A-', 'A+') THEN 4
                WHEN cg.final_grade IN ('B', 'B-', 'B+') THEN 3
                WHEN cg.final_grade IN ('C', 'C-', 'C+') THEN 2
                WHEN cg.final_grade IN ('D', 'D-', 'D+') THEN 1
                WHEN cg.final_grade = 'F' THEN 0
            END
        ) AS avg_grade
    FROM students s
    JOIN course_grades cg ON s.student_id = cg.student_id
    GROUP BY s.student_id
),
stats AS (
    SELECT
        s.major,
        AVG(sh.hours_solo) AS avg_solo,
        AVG(sh.hours_group) AS avg_group,
        AVG(sh.hours_tutoring) AS avg_tutoring,
        AVG(sh.hours_solo * sh.hours_solo) AS avg_solo_sq,
        AVG(sh.hours_group * sh.hours_group) AS avg_group_sq,
        AVG(sh.hours_tutoring * sh.hours_tutoring) AS avg_tutor_sq
    FROM students AS s
    JOIN study_habits AS sh ON s.student_id = sh.student_id
    GROUP BY s.major
)
SELECT
    major,
    (avg_solo_sq - avg_solo * avg_solo) AS var_solo,
    (avg_group_sq - avg_group * avg_group) AS var_group,
    (avg_tutor_sq - avg_tutoring * avg_tutoring) AS var_tutoring
FROM stats
ORDER BY (var_solo + var_group + var_tutoring) DESC;
```

	major	var_solo	var_group	var_tutoring
1	Computer Science	4.77609262182069	2.44254676669528	0.678170103464145
2	Biology	4.42205985738225	1.34743898292958	0.686756635956771
3	Psychology	3.53974313208001	1.69937148616001	0.57103364722686
4	Business	2.74919788455507	1.82046393866692	0.972446850657496
5	Sociology	2.87245780212349	1.75875702960292	0.808684317167542
6	Art	2.53080357826023	1.92990977307496	0.793412171148404

This query shows the variation of students in each major in what they like to study like. So we can see that computer science majors have the highest variation of study methods, and art majors have the lowest variation .

- <u>Do students who work more than they study get lower grades?</u>

```sql
-- Do students who work more than they study get lower grades?
SELECT
AVG(
    CASE
        WHEN cg.final_grade IN ('A', 'A-', 'A+') THEN 4
        WHEN cg.final_grade IN ('B', 'B-', 'B+') THEN 3
        WHEN cg.final_grade IN ('C', 'C-', 'C+') THEN 2
        WHEN cg.final_grade IN ('D', 'D-', 'D+') THEN 1
        WHEN cg.final_grade = 'F' THEN 0
    END
) AS avg_grade
FROM course_grades AS cg
INNER JOIN study_habits AS sh
USING (student_id)
INNER JOIN activities AS a
USING (student_id)
WHERE sh.hours_solo + sh.hours_tutoring +sh.hours_group < a.job_hours;
```

	avg_grade
1	2.67326732673267

This query shows the students who have more work hours than total study hours. For this query, we had to join 3 tables together to get the GPA.

```sql
-- Do students who work more than they study get lower grades?
SELECT
AVG(
    CASE
        WHEN cg.final_grade IN ('A', 'A-', 'A+') THEN 4
        WHEN cg.final_grade IN ('B', 'B-', 'B+') THEN 3
        WHEN cg.final_grade IN ('C', 'C-', 'C+') THEN 2
        WHEN cg.final_grade IN ('D', 'D-', 'D+') THEN 1
        WHEN cg.final_grade = 'F' THEN 0
    END
) AS avg_grade
FROM course_grades AS cg
INNER JOIN study_habits AS sh
USING (student_id)
INNER JOIN activities AS a
USING (student_id)
WHERE sh.hours_solo + sh.hours_tutoring +sh.hours_group > a.job_hours;
```

	avg_grade
1	2.93493150684932

This query shows the students who have more study hours than work hours. For this query, we had to join 3 tables together to get the GPA.

To answer this question, we can see that the students who have more study time than work hours have better grades by about 0.3. This I much less of a difference than I was expecting